Telecom Revenue Optimization Report

Generated: March 04, 2026

Data Source: IBM Telco Customer Churn (Kaggle)

Customers Analyzed: 7,043

Powered by Smart Retention Engine, NBO Predictions, and ML Explainability

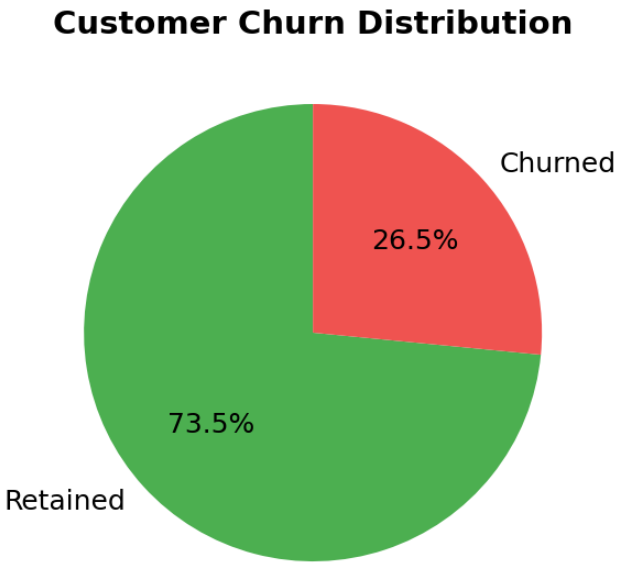
1. Executive Summary

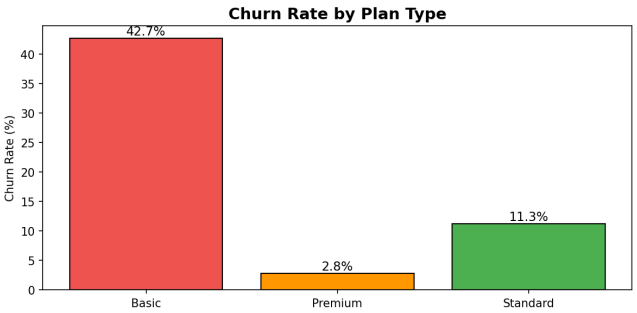
* Total Customers: 7,043
* Monthly Revenue: $456,116.60
* Average ARPU: $64.76
* Current Churn Rate: 26.5%
* Revenue at Risk (from churned): $139,130.85/month
* Average Satisfaction: 7.7/10

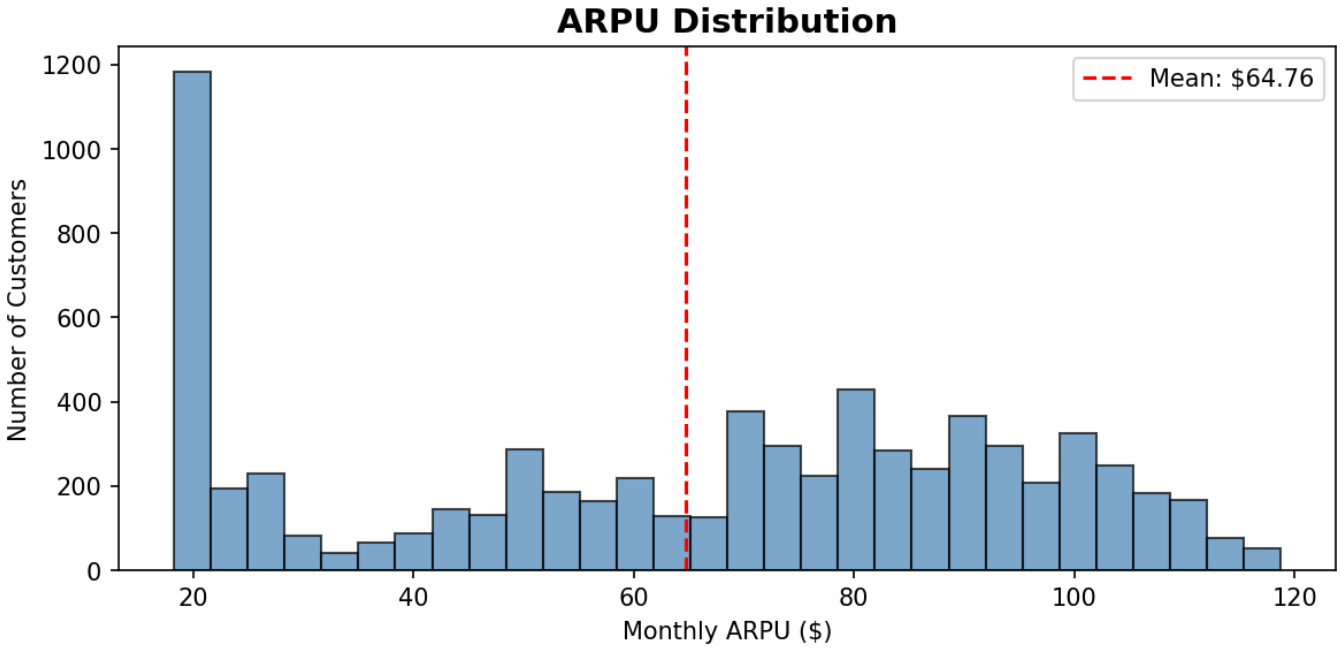
Retention Engine Results:

* At-risk customers: 5,171 (73.4%)
* Average Health Score: 64.3/100
* Total intervention cost: $73,685
* Expected revenue saved: $319,196

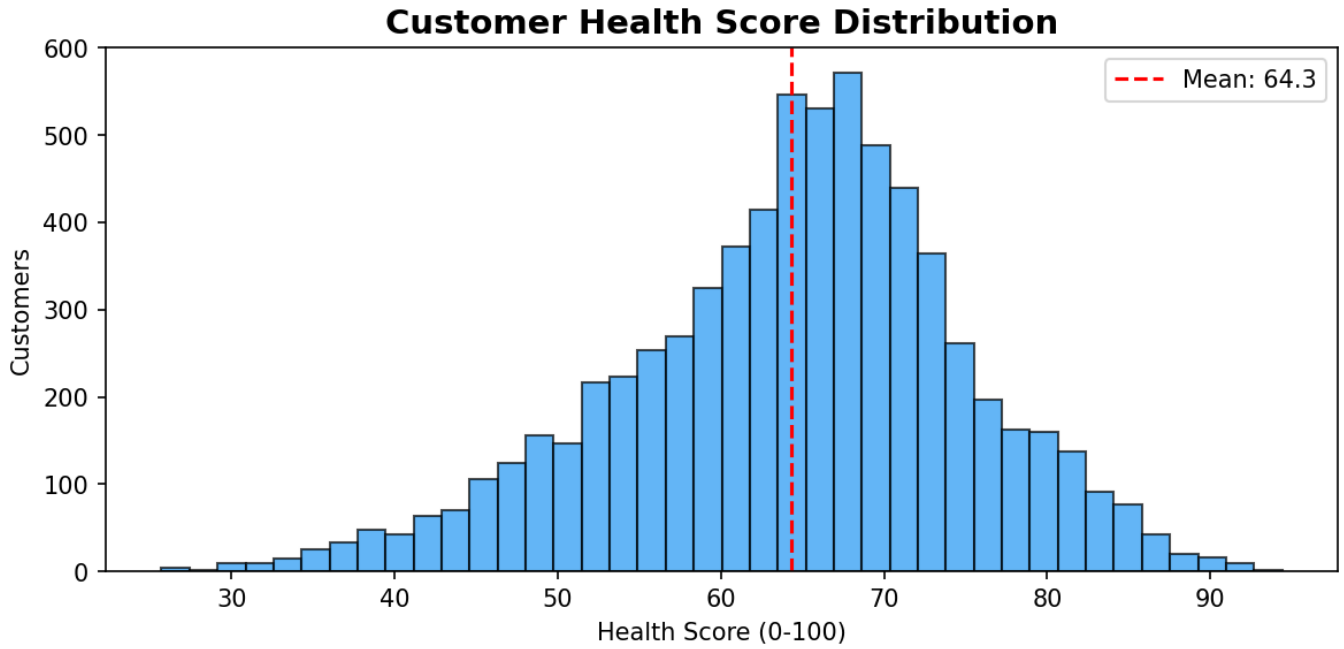
2. Customer Analysis

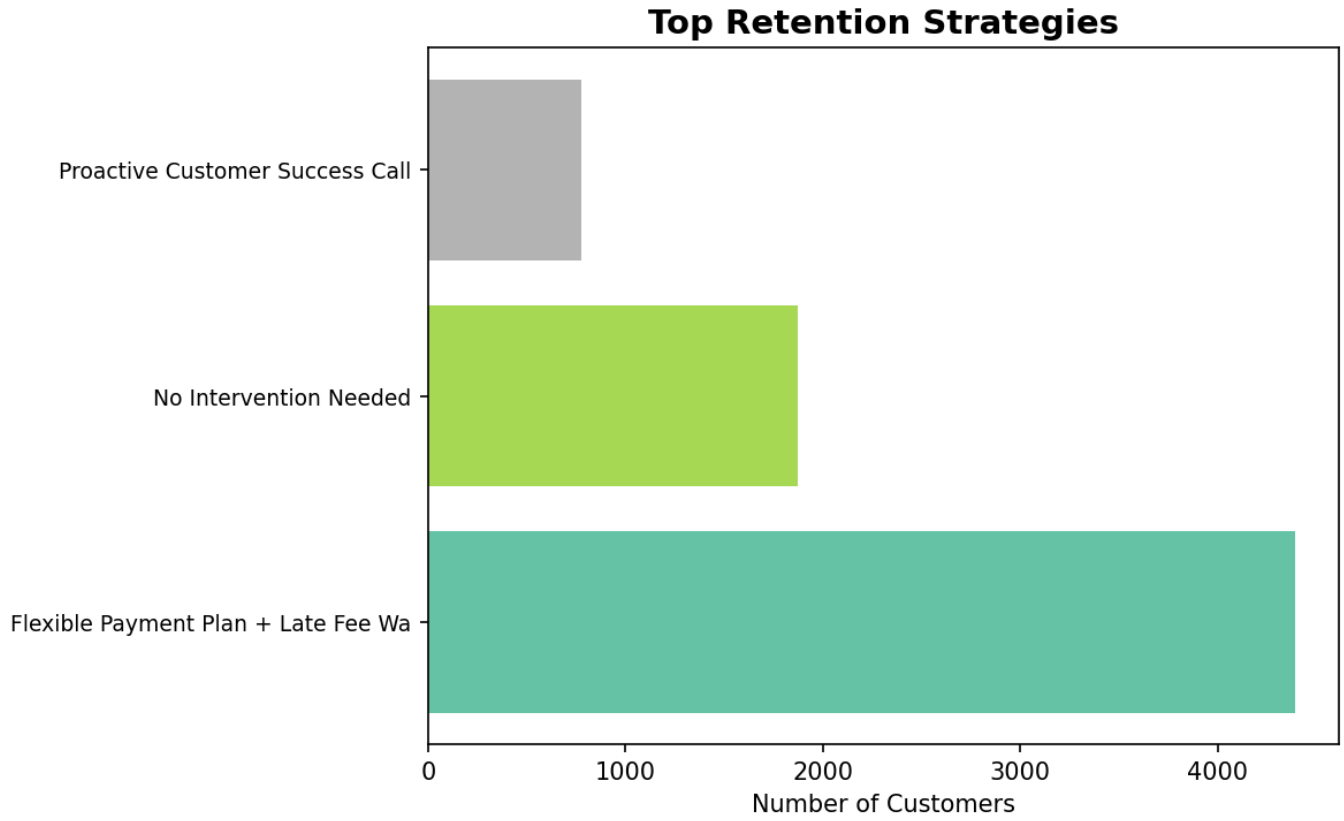






3. Retention Strategy Analysis





4. Next Best Offer Analysis

* Total expected annual revenue from NBO: $286,821
* Average conversion probability: 26.6%
* Customers with offers: 7,034

Top Offers by Revenue Potential:

Upgrade to Premium Plan: $69,296
Data Boost Package: $49,053
Upgrade to Standard Plan: $47,401
Add Streaming Bundle: $38,433
Add Premium Tech Support: $37,771
Add Online Security & Backup: $28,725
Loyalty Rewards Bundle: $16,142

5. Recommendations

1. Deploy the Smart Retention Engine to proactively identify and retain at-risk customers.
2. Implement the Next Best Offer system to drive upsell/cross-sell revenue growth.
3. Focus retention budget on high-value customers with the highest expected ROI.
4. Address the primary churn drivers (low engagement, competitor risk, price sensitivity).
5. Monitor the Customer Health Score dashboard weekly for early warning signals.
6. Run A/B tests on retention strategies to optimize intervention effectiveness.
7. Invest in plan upgrades for Basic-plan customers, who show the highest churn risk.